STUDENT AGREEMENT

between

ETH Zurich («ETH Zürich»)
Rämistrasse 101, 8092 Zurich, represented by Prof. Christian Holz (SIPLAB: Sensing, Interaction & Perception Lab), hereafter referred to as "**Hosting Unit**"

and

Tim Goppelsroeder
hereafter referred to as "**Student**"

concerning the Student's participation in a research project in the framework of his/her master, semester or bachelor thesis work.

1. Preamble

The Hosting Unit is conducting or has the intention to conduct a research project, as described in Appendix A ("**Project**"). The Hosting Unit has an interest in being able to use all results of the Project for publications, as a basis for further research and/or for granting rights in the results to third parties.

The Project or a part of the Project may be performed by one or more master or bachelor students. The Student has an interest in participating in the performance of the Project in his/her bachelor or master thesis work. The Hosting Unit is willing to allow the Student to participate in the Project subject to the conditions of this Agreement, e.g. concerning the results of the Student in the Project ("**Student Results**").

Therefore, the Parties will enter into the following agreement ("**Agreement**"):

2. 1. Assignment of Results

1.1 With the exception of copyrights in scientific publications including his/her thesis, the Student herewith assigns to ETH Zurich irrevocably and at no cost all of his/her rights, title and interest in and to the Student Results including but not limited to rights in inventions, software and other copyright protected work, but excluding his/her moral rights attached ("Urheberpersönlichkeitsrechte"). For software, to the extent permitted by the applicable law, the Student renounces his/her right to be named as author. For the assigned copyrights, the right of first publication and the right to modify for such works vest in ETH Zurich. In exchange and where applicable on a case-by-case basis, the Student will be treated as an "inventor" or a "creator of a copyrighted work" in the sense of the "Guidelines for the Financial Exploitation of Research Results at ETH Zurich", RSETHZ 440.4.

1.2 Where assignment is not permitted by law, an exclusive, worldwide, irrevocable, perpetual, unlimited, royalty-free and sublicensable license for the use of the Student Results is hereby granted to ETH Zurich instead.

1.3 The Student shall prepare and maintain adequate records of his/her work at ETH Zurich and will provide to ETH Zurich copies of such records upon request. The Student further agrees to timely provide any signatures to enable ETH Zurich to perfect any of its rights in the Student Results.

3. 2. Confidentiality

2.1 ETH Zurich does not wish to receive any proprietary or confidential information under this Agreement. The Student therefore agrees and acknowledges that, except as provided otherwise in writing, any information and data provided by Student to ETH Zurich is of a non-confidential nature and can be used and disposed of by ETH Zurich free from any constraints.

2.2 In the course of the Student's stay at ETH Zurich or his/her access to ETH Zurich infrastructure, the Student might be exposed to confidential information belonging to ETH Zurich or a third party ("**Confidential Information**"). The Student Results shall be deemed ETH Zurich's Confidential

Information. The Student shall treat all Confidential Information as confidential and shall use it solely for the Project. Except as provided in Section 3, the Student shall not disclose Confidential Information to any third party. The Student shall abide by such non-disclosure obligations for the duration ETH Zurich has legitimate interest in keeping the Confidential Information in secrecy. Except for a copy of the thesis, Student shall, upon ETH Zurich's request, either return or destroy any Confidential Information and any digital and physical copies containing Confidential Information in the Student's possession.

4. 3. Publication

The Student acknowledges that any scientific publication or other disclosure of the Student Results to third parties by any means might jeopardize the interests of ETH Zurich. The Student agrees to request from the Hosting Unit prior approval for any publication of Student Results, such approval not to be unreasonably withheld or delayed. The authorship will be defined according to the criteria of Article 14 "Author Information" of the Guidelines for Research Integrity and Good Scientific Practice at the ETH Zurich (RSETHZ 414).

5. 4. Term and Termination

4.1 The Agreement enters into force upon signature by ETH Zurich and the Student and shall have effect from the earlier of 1) Student's start of participation in the Project or 2) Student's access to ETH Zurich's infrastructure or Confidential Information. The Agreement will continue in full force and effect until the end of the Student's participation in the Project.

4.2 The provisions which, by their nature are intended to survive the expiry or termination of the Agreement shall continue to apply.

6. 5. Miscellaneous

5.1 The Student affirms that he/she did not enter into a commitment with any third party that is inconsistent with any obligations entered in this Agreement.

5.2 The Student and ETH Zurich agree that nothing contained in this Agreement shall imply any partnership or employer-employee-relationship between Student and ETH Zurich.

5.3 If individual provisions of this Agreement are invalid or the fulfilment thereof is impossible, the validity of the remaining parts of the Agreement shall not be affected. Amendments and additions to this Agreement are only valid if made in writing and legally signed by the Student and ETH Zurich.

5.4 This Agreement shall be construed and exclusively governed by the laws of Switzerland, without reference to its conflict of laws principles. The sole place of jurisdiction for any dispute arising from, or in connection with, the contract shall be exclusively the courts of the city of Zurich.

5.5 Appendix A forms an integral part of this Agreement. In the case of deviations between the Appendix A and the Agreement, the Agreement shall prevail.

ETH Zurich	Zurich, ………………….	**Student**	Zurich, ………………….
Prof. Christian Holz		Tim Goppelsroeder	

Appendix A: Description of Project

ETH zürich

sensing, interaction & perception lab

Semester thesis

Discovering Reward Functions for Realistic Biomechanical Simulation

Introduction

Biomechanical models have human physical structures and constraints, offering the potential to generate and simulate realistic human motions in a wide range of tasks. A common approach to control such models is via reinforcement learning (RL), which involves acquiring control policies through trial-and-error interactions within a simulated environment. However, a significant challenge is motor redundancy, referring to the phenomenon where multiple combinations of muscle activations or joint motions can result in the same overall movement. For example, when using a trackpad, multiple hand or finger postures could achieve the same cursor movement, but the most human-like and efficient interaction typically involves the use of the index fingertip. This redundancy complicates policy learning, often leading to successful, yet unrealistic motion behaviors.

To address motor redundancy, the design of the reward function becomes a central challenge. Standard RL formulations typically emphasize task completion (for example, reaching a target or moving object to the right target position, etc) without considering whether the motion itself (i.e., the sequence of actions) resembles human biomechanics. As a result, policies may exploit shortcuts or unnatural strategies that technically solve the task but deviate from realistic human movement. For instance, an agent controlling a simulated arm might flail its joints erratically to touch a target, or a walking model may adopt energy-inefficient gaits that would be implausible for humans. Crafting a reward function that captures both task success and biomechanical plausibility is thus crucial, yet it remains difficult due to the high dimensionality of the body, the variability of human strategies, and the lack of explicit metrics for "realism."

This project aims to systematically explore and design reward functions that encourage both task success and biomechanical realism. Inspired by recent advances such as Project Eureka (see: https://eureka-research.github.io/), the approach will leverage large language models (LLMs) to automatically generate, refine, and evaluate candidate reward functions. Unlike Eureka, which relies on large proprietary models, this project will investigate using local, open-source models (e.g., LLaMA) to enable iterative, controllable, and resource-efficient experimentation. The student will integrate these LLM-driven proposals with biomechanical criteria, such as energy efficiency, smoothness, and similarity to human movement patterns from human demonstration.

Specifically, we will focus on these milestones: (1) begin with simpler RL tasks in OpenAI Gym to validate the LLM-driven reward generation pipeline, (2) extend to reaching tasks with joint motor control to test how well the proposed rewards address motor redundancy, and (3) progressively scale to more complex biomechanical simulations (e.g., locomotion or grasping) where realism is harder to capture. At each stage, the student will benchmark policies trained with LLM-generated reward functions against baseline manually created reward functions.

Task description

The goal of this thesis is

1. Read the state-of-the-art reward function discovery paper, and concretely design a feasible pipeline for our goal.
2. Prior preparation for the project:
    a. Install tools for physics simulation, including Mujoco [https://mujoco.org/] and Myosuite [https://sites.google.com/view/myosuite]. Then, implement (run) the most basic interactions to get familiar with the setup.
    b. Install state-of-the-art RL libraries, such as stable-baselines [https://github.com/hill-a/stable-baselines].
    c. Install OpenAI Gym and get familiar with it.
3. Implement the method(s) based on the previous preparations:
    a. First we will install Llama (or other suitable models) on our lab server.
    b. Use the LLM to automatically propose, refine, and evaluate candidate reward functions for progressively more complex RL tasks.
    c. Benchmark the discovered reward functions against baselines (task-only rewards and manually crafted rewards) on both simple Gym tasks.
    d. Document findings and iterate on the pipeline to improve realism, efficiency, and reproducibility.
4. Deploy the methods on biomechanical simulations, from simpler tasks to more complex tasks.
5. Finally, we will need to report the final performance, in both performance metrics and the perceived realism gathered from human viewers.

Submission

1. A written thesis (two-column layout, template provided, writing typically takes 2 weeks minimum). We recommend adding references and comments on how they relate to your project as you discover them.
2. A clean repository of all your code (e.g., a documented Jupyter Notebook or Github repository), including results and graphs (e.g., as part of the Notebook or as markdown files in the repo).
3. An oral presentation with an interactive Q&A for completing this thesis.

Grading scale

The thesis will be graded using the following scale:

6.0 Results and thesis quality exceptionally exceed the expectations laid out above (e.g., equivalent to publication quality at top-tier academic venues)

5.5 Results and thesis quality significantly exceed expectations
5.0 Results and thesis meet expectations
4.5 Thesis partially meets expectations, minor deficits
4.0 Thesis meets minimal quality requirements; it has major deficits and it is clearly below expectations

Dr. Yi-Chi Liao

Prof. Dr. Christian Holz